

Introduction to Diffusion Probabilistic Models

Acknowledgement:

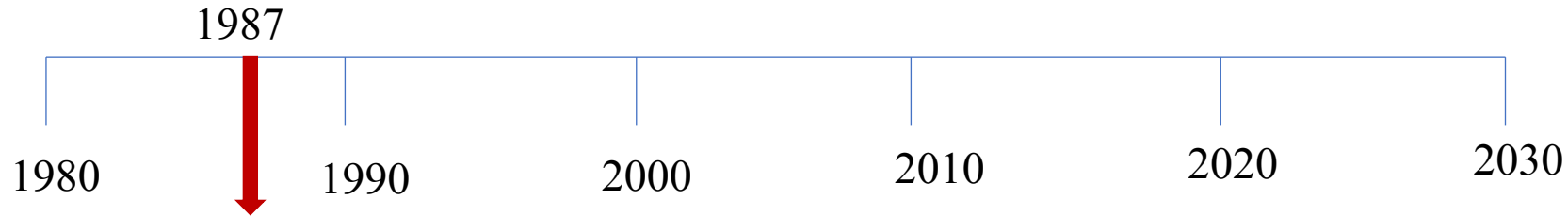
Certain concepts in this PPT were inspired by Julia Turc's educational video on Diffusion Models available on YouTube.

<https://www.youtube.com/watch?v=R0uMcXsfo2o>

Content

- Evolution from Autoencoders to Diffusion Models
- Introduction to Diffusion Model.
- Need for Diffusion Model
- Forward and Reverse Diffusion Process
- Training and Inference Pipeline
- Introduction to Markov Chain

From VAEs/GANS to Diffusion Models



Autoencoders were introduced.

Ex: Similar to ZIP compression

Used for image generation
Reconstruct the input from a compressed latent representation

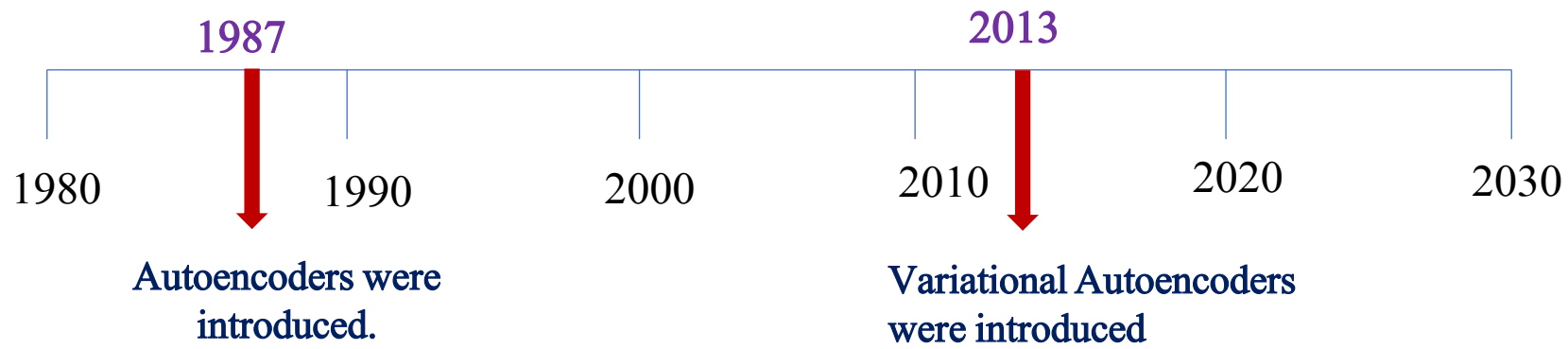
Applications

- ✓ Remove unwanted noise from data and improve quality.
- ✓ Identifies unusual patterns or anomalies in data.
- ✓ Extracts important features for better model performance



Image Restoration

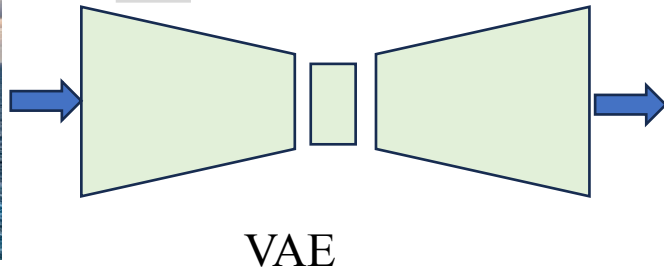
Limitation:
Traditional autoencoders cannot generate realistic new images.

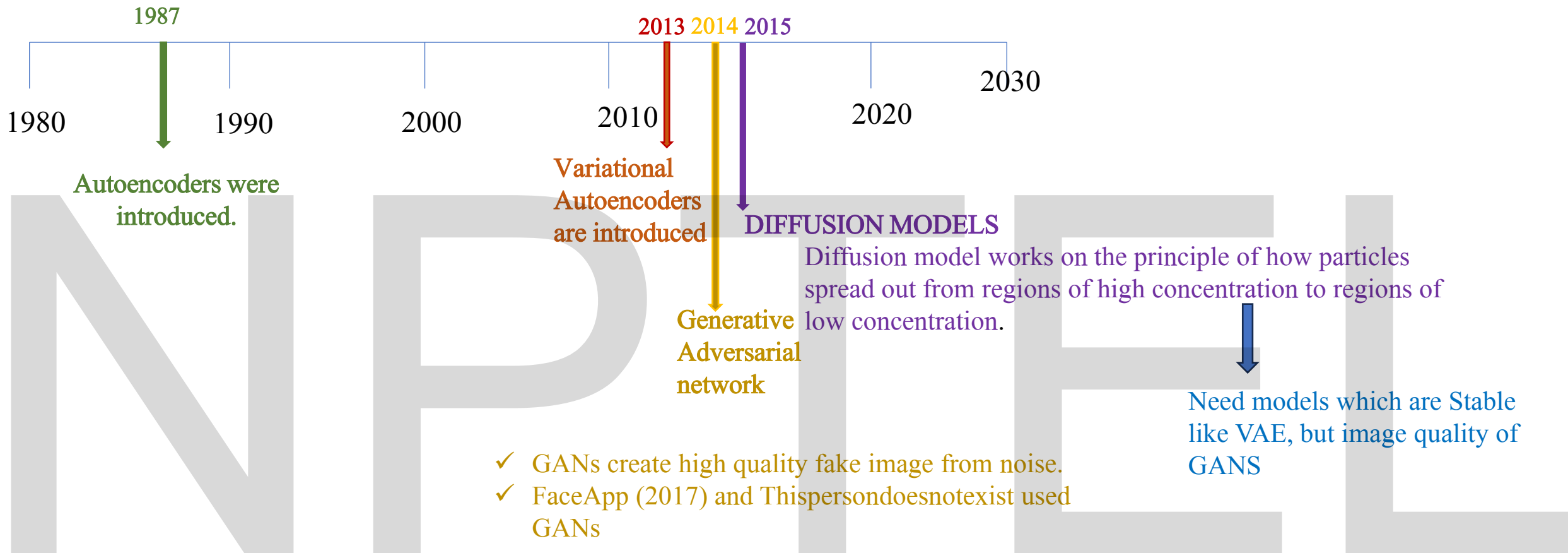


- ✓ VAEs stored the internal representation of the image in terms of probability distribution.
- ✓ Represents the input image in terms of mean and variance (latent Gaussian Distribution)

Limitation!!
Often Generated Images are blurry

Can generate new images





- ✓ GANs create high quality fake image from noise.
- ✓ FaceApp (2017) and Thispersondoesnotexist used GANs

Limitation!!
Training instability
MODE COLLAPSE

Diffusion Model

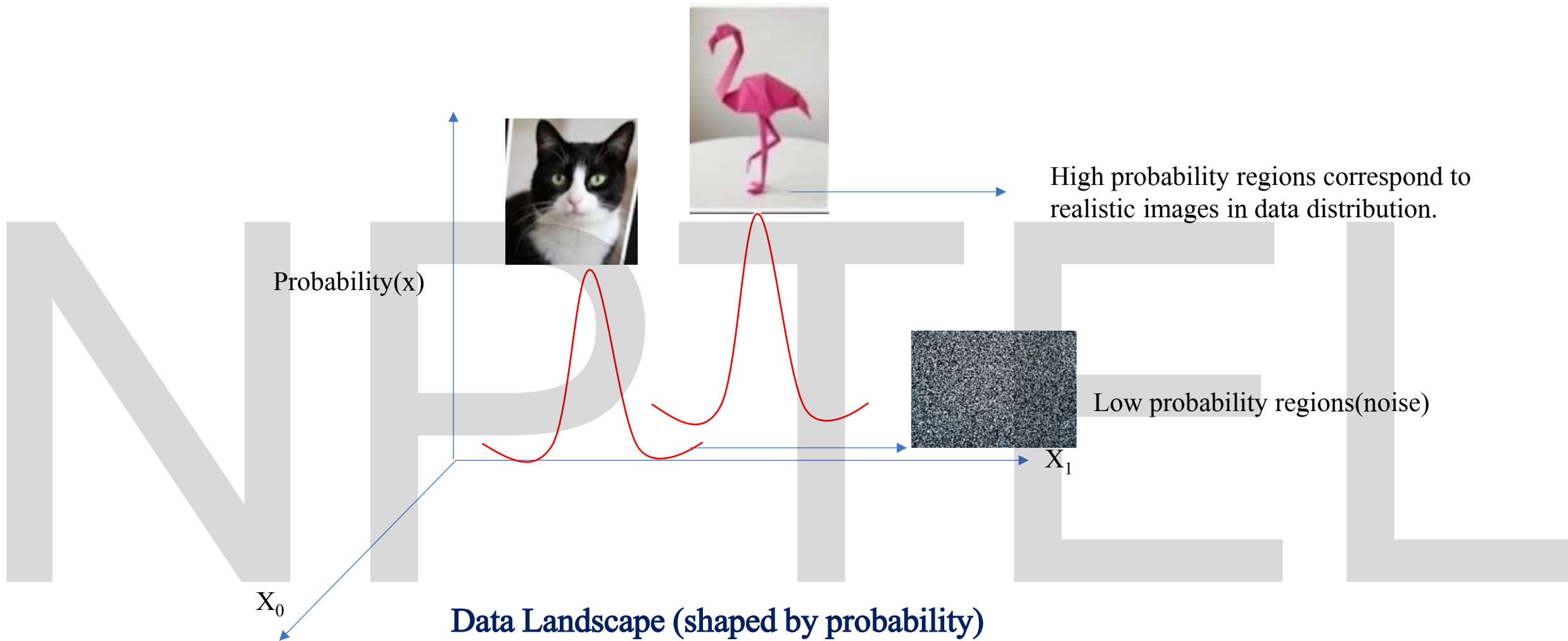
In 2015, researchers connected machine learning with concepts from nonequilibrium thermodynamics-Branch of physics that discusses how ink spreads in water or how perfume spreads in air.

Physics of Particle
motion

Diffusion Model

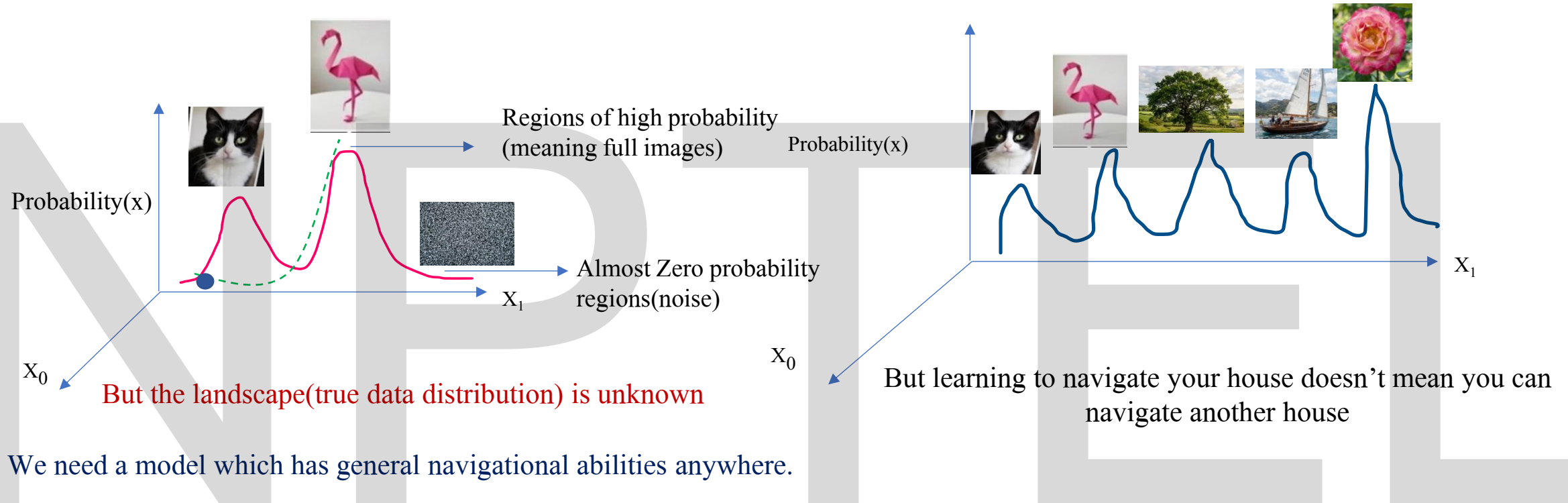
Generative AI





- ✓ In pixel space, each pixel channel will have its own axis.
- ✓ For simplicity we take here only two axes

How do we Visualize Data Generation



- We need a model which has general navigational abilities anywhere.
- It should be able to make local decision at each step moving towards the region of higher probability compared to its current position.



Physics in Diffusion Models

The gradual exploration can be modeled using a time-varying probability distribution.

Instead of $p(x)$ we will work with $p_t(x)$

$$p_t(x) = p(x|t)$$

Probability of data x at time t

- ✓ New concept in machine learning.
- ✓ That is where Physics came into picture where time dependent fields are common.



Concentration changes over space and time

Why were Diffusion Models Introduced

Core Problem of existing Generative Model

Most generative models fall into one of the categories

1. **Tractable** but not **expressive** → Cannot model complex real-world data (images, speech)



A model is tractable if we can easily:

- ✓ Compute probabilities $p(x)$
- ✓ Train the model (optimize parameters)
- ✓ Sample from it (generate data)

2. **Expressive but intractable.**

We can't

- Evaluate how good it is
- Train it efficiently
- Generate high-quality samples efficiently

Ref:

Jascha Sohl-Dickstein et al., *Deep Unsupervised Learning using Nonequilibrium Thermodynamics*, arXiv:1503.03585

Why were Diffusion Probabilistic Models Introduced

We want a model that is BOTH:

- ✓ Highly flexible
- ✓ Computationally tractable

Diffusion models solve this by breaking a hard problem into many easy steps:

Forward Process (known, fixed)

- ✓ Gradually add noise to data
- ✓ Convert complex data $\rightarrow$ simple Gaussian
- ✓ $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow \dots \rightarrow x_T \sim N(0, I)$
- ✓ This is **easy and tractable**

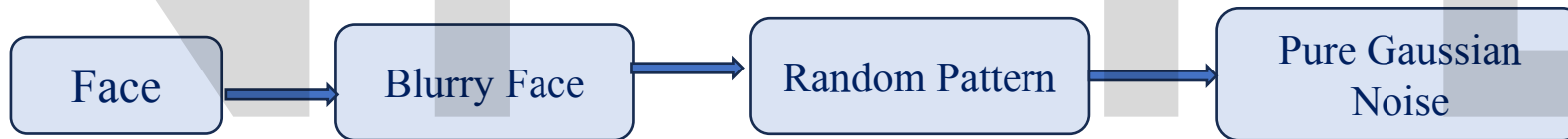
Reverse Process

- ✓ Learn to reverse noise step-by-step
- ✓ $x_T \rightarrow x_{T-1} \rightarrow \dots \rightarrow x_0$
- ✓ Each denoising step becomes easier to learn.

Diffusion Model

- ✓ Diffusion models are a type of generative AI that generate images, audio, and video from random noise.
- ✓ A controlled process gradually transforms structured data into noise.
- ✓ The model then learns the reverse denoising process.
- ✓ Image : Signal with all of structured information
- ✓ Noise: Unstructured data

Forward Diffusion

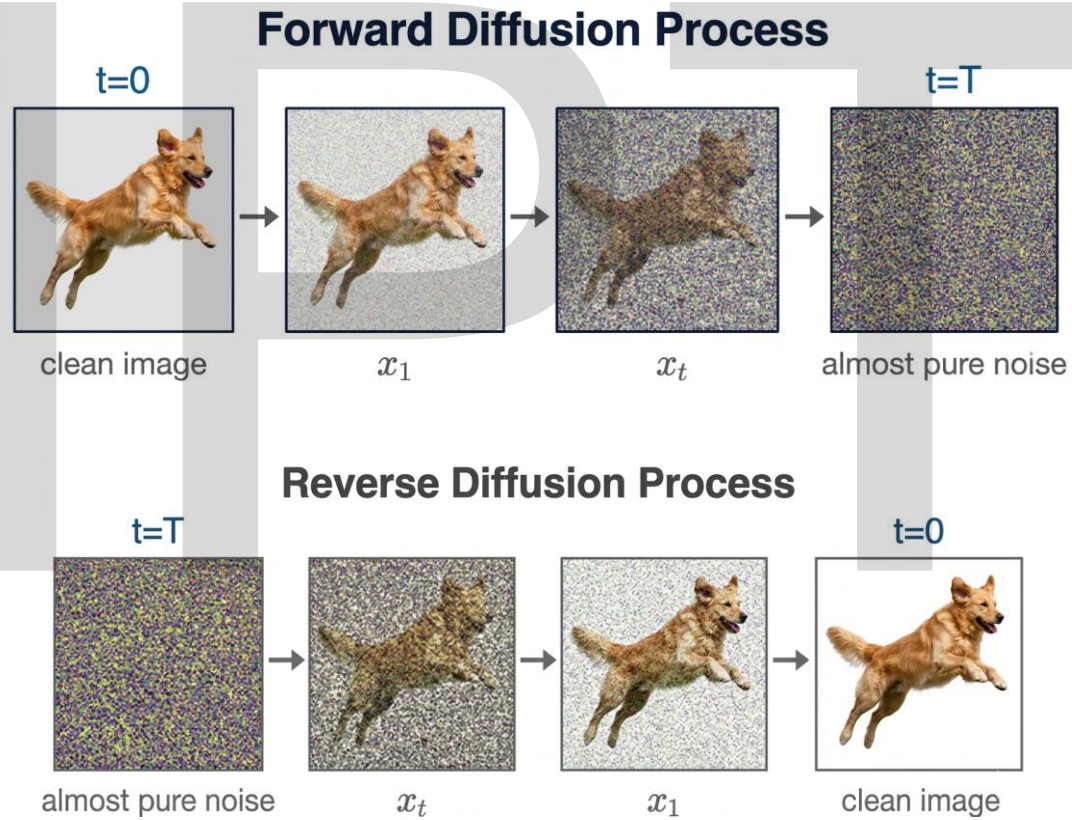


Everything becomes a simple distribution like:
$$x_T \sim \mathcal{N}(0, I)$$

Core Idea of Diffusion Model

Learning $p(x)$ directly is hard

But learning $p(x_{t-1}|x_t)$ which is much easier.



In Forward process only noise is added

In Reverse process the actual learning happens

Training and Inference Pipeline

Training pipeline

Take real image
 x_0

Add noise to get
 x_t

Give noisy image to
neural network

Neural Network
predicts the
added noise

Compute
denoising loss
between true
and predicted
noise.

Update
weights

Inference(Sampling) pipeline

Generation starts from
pure random noise:
 $x_T \sim \mathcal{N}(0, I)$

Trained model repeatedly
denoises:
 $x_T \rightarrow x_{T-1} \rightarrow x_{T-2} \rightarrow \dots \rightarrow x_0$

Repeated denoising
gradually generates a
realistic image.

Introduction to Markov Chain

A **Markov chain** describes a system that transitions between different situations, called states.

Probability of moving to the next state depends only on the current state and not on the sequence of states that came before it.



Mathematically:

$$P(x_t | x_{t-1}, x_{t-2}, \dots, x_0) = P(x_t | x_{t-1})$$

To generate x_t , only x_{t-1} is needed

Forward Noising Process

Reverse Denoising Process

Both Modeled as Markov chains.

NPTEL

Summary

Next Session

Mathematics of Diffusion Model